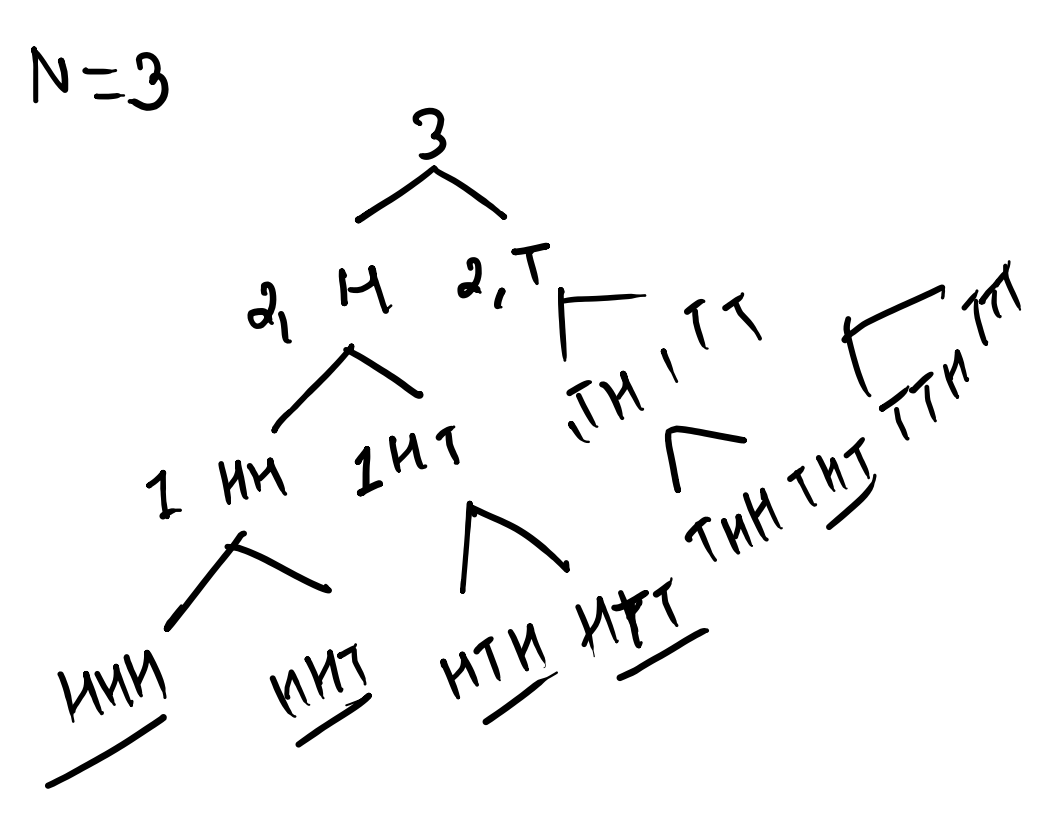
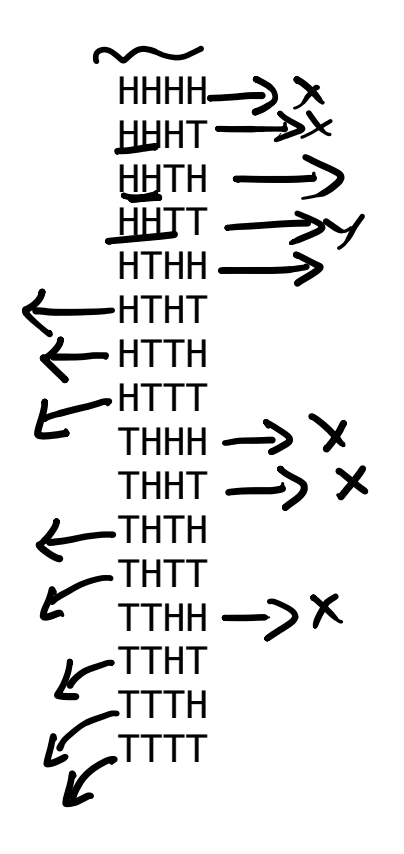
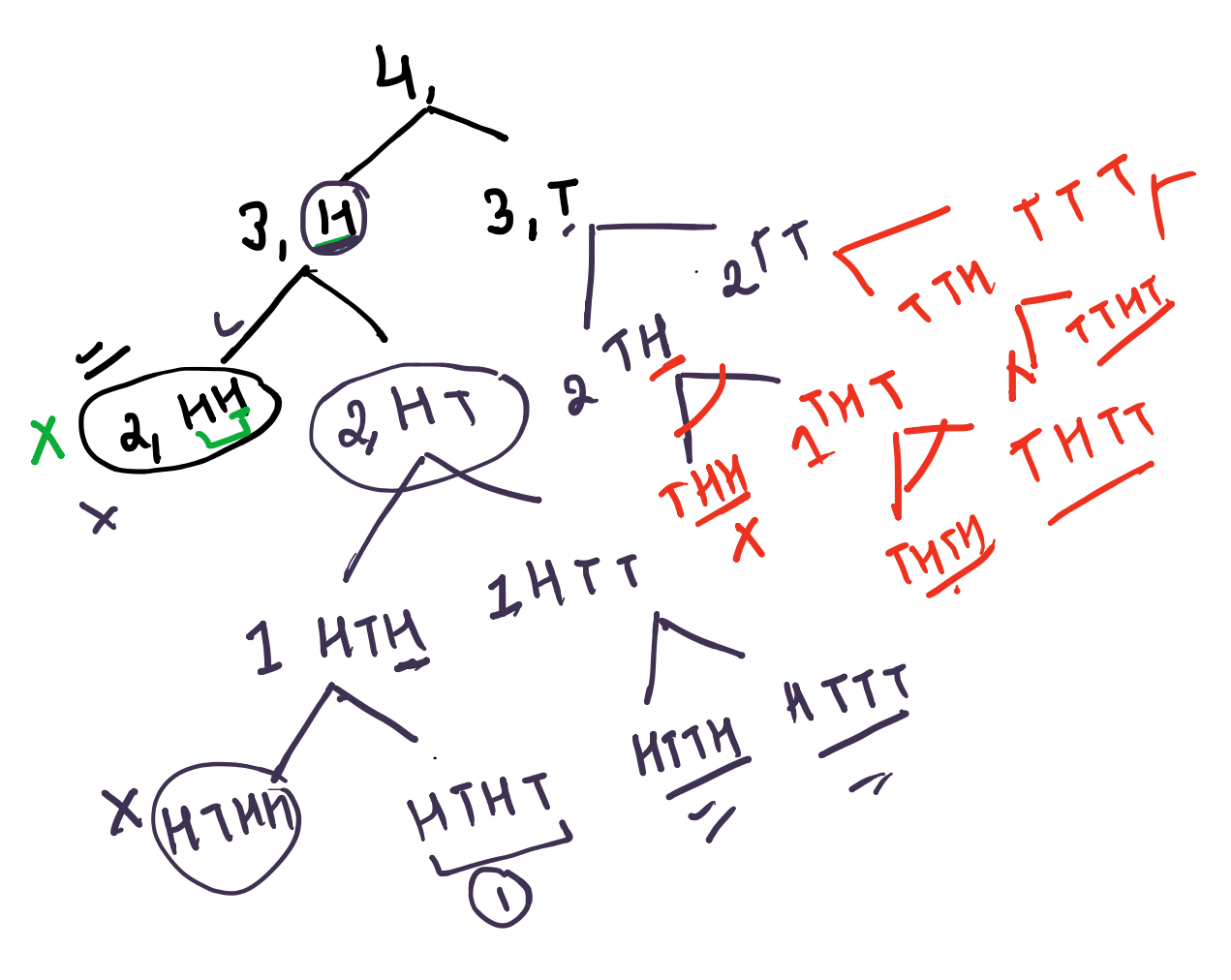
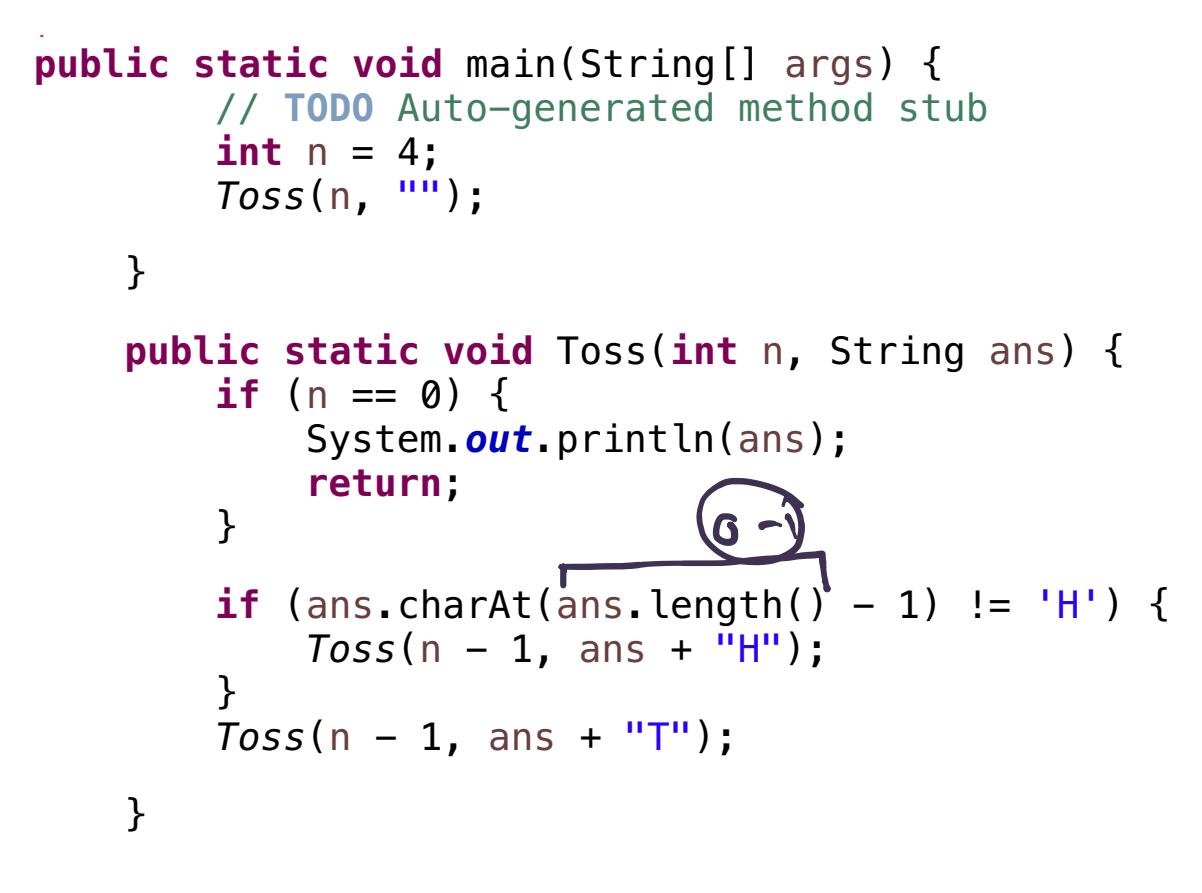
# Lec-20

19/06/26 7:10 PM

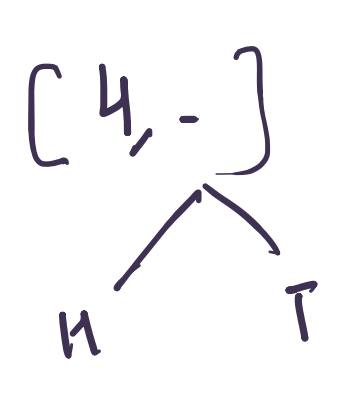

N=3
3
2, H
2, T
1 HH
1 HT
TH
TT
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT

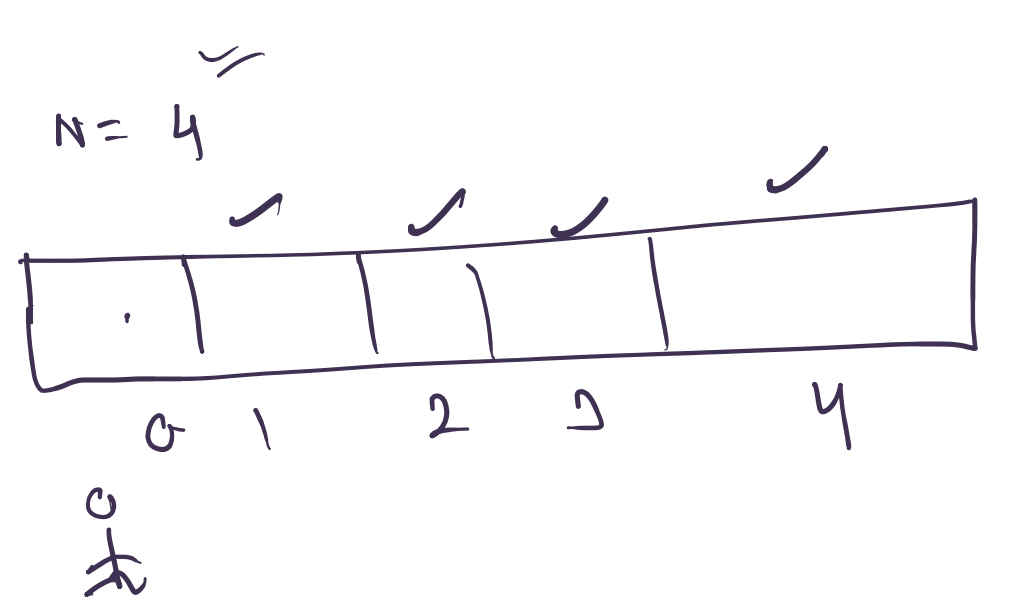

HHHH
HHHT
HHTH
HHTT
HTHH
HTHT
HTTH
HTTT
THHH
THHT
THTH
THTT
TTHH
TTHT
TTTH
TTTT


4,
3, H
3, T
2, HH
2 HT
2 TH
2 TT
1 HTH
1 HTT
HTHH
HTHT
HTTH
HTTT
THH
THT
TTH
TTT
THTH
TTHT
THTT



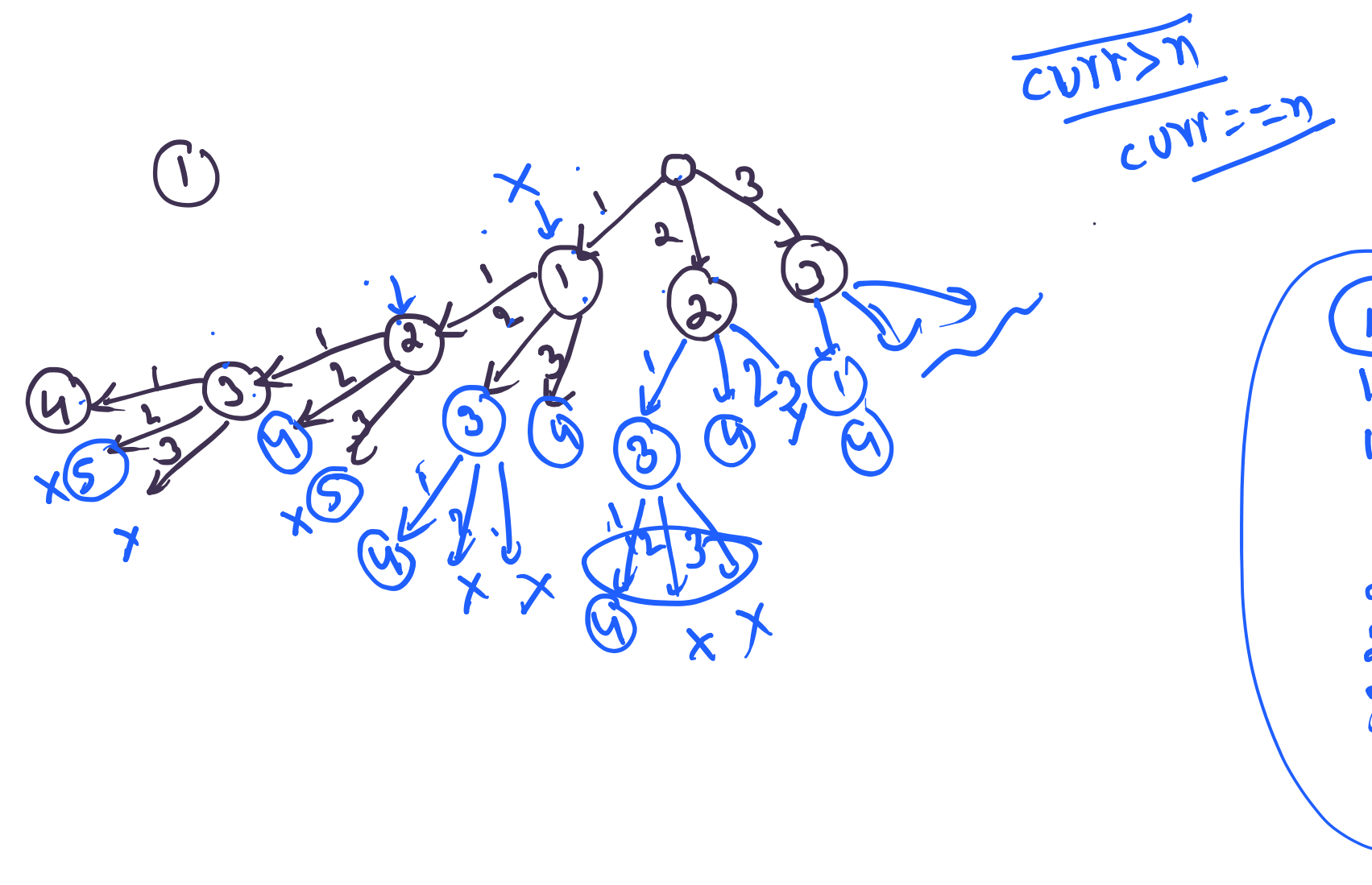
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 4;
        Toss(n, "");

    }

    public static void Toss(int n, String ans) {
        if (n == 0) {
            System.out.println(ans);
            return;
        }

        if (ans.charAt(ans.length() - 1) != 'H') {
            Toss(n - 1, ans + "H");
        }
        Toss(n - 1, ans + "T");

    }
```





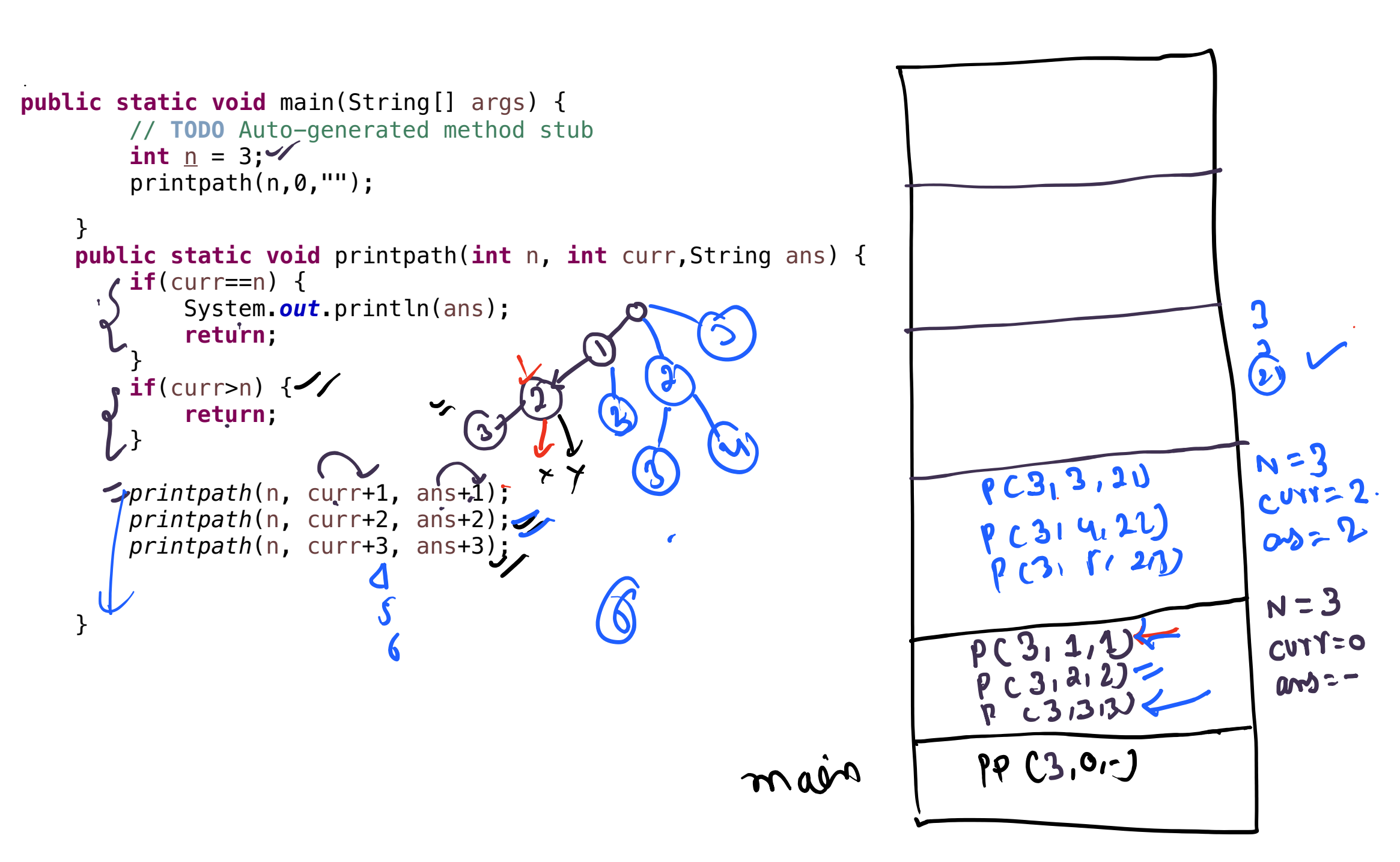
1,2,3,4,5,6

dice=1,2,3


curr>n
curr==n
1

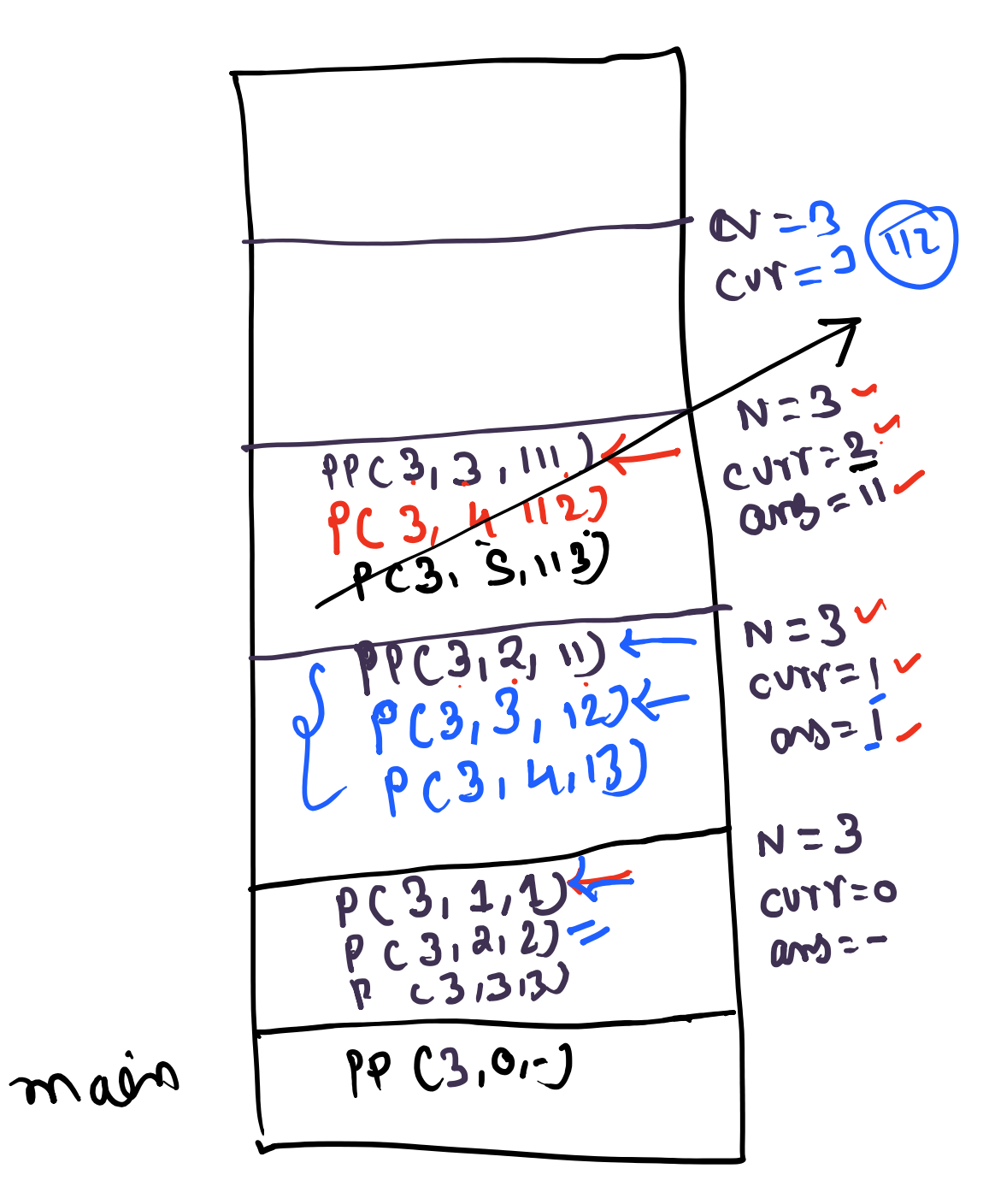
1111
112
121
211
22
13
31

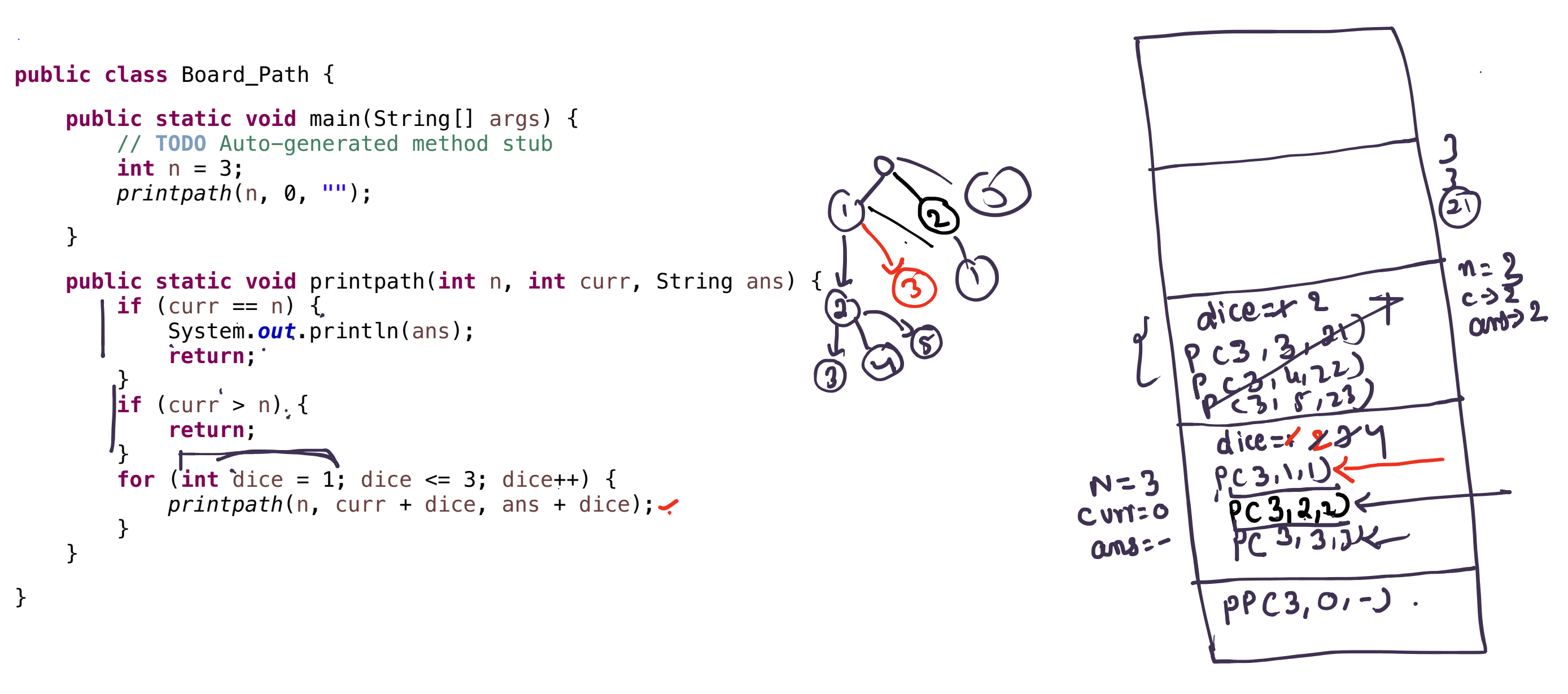
1111
112
121
13
211
22
31



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 3;
        printpath(n,0,"");

    }
    public static void printpath(int n, int curr,String ans) {
        if(curr==n) {
            System.out.println(ans);
            return;
        }
        if(curr>n) {
            return;
        }

        printpath(n, curr+1, ans+1);
        printpath(n, curr+2, ans+2);
        printpath(n, curr+3, ans+3);

    }
```

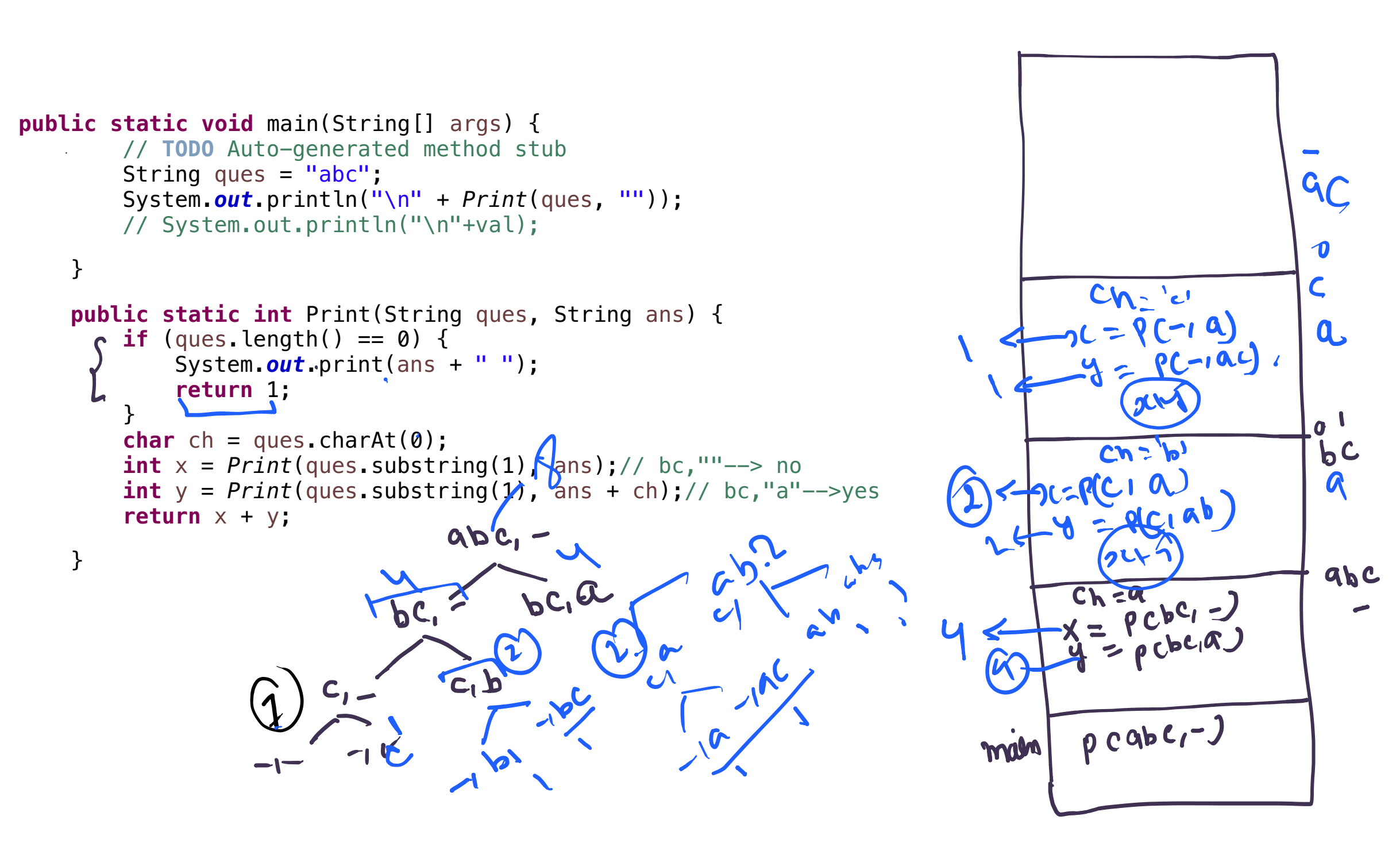
4
5
6
PC3,3,21)
PC3,4,22)
PC3,5,23)
PC3,1,1)
PC3,2,2)
PC3,3,3)
PP(3,0,-)
main
N=3
curr=2
ans=2
N=3
curr=0
ans=-


N=3
curr=3
112
PPC3,3,111)
PC3,4,112)
PC3,5,113)
N=3
curr=2
ans=11
PPC3,2,11)
PC3,3,12)
PC3,4,13)
N=3
curr=1
ans=1
PC3,1,1)
PC3,2,2)
PC3,3,3)
N=3
curr=0
ans=-
PP(3,0,-)
main



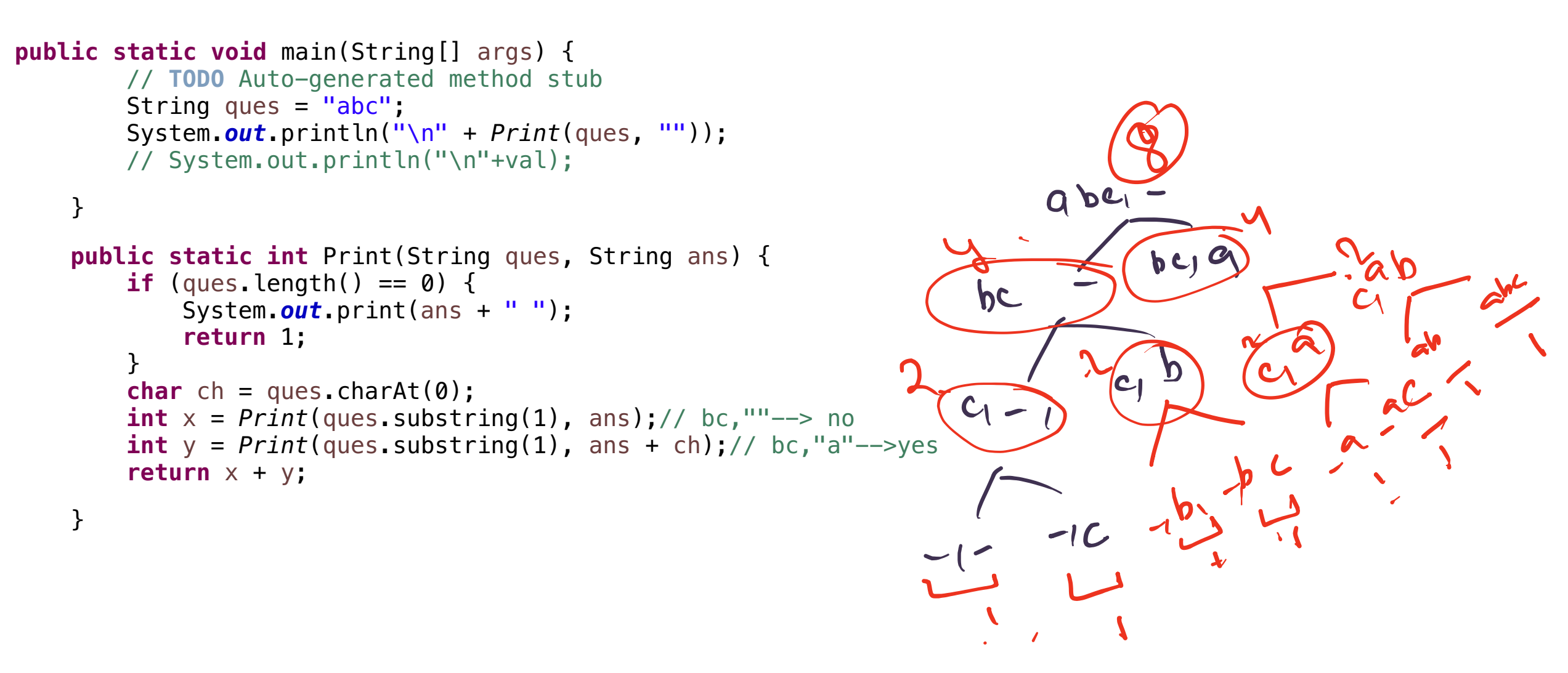
```
public class Board_Path {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 3;
        printpath(n, 0, "");

    }

    public static void printpath(int n, int curr, String ans) {
        if (curr == n) {
            System.out.println(ans);
            return;
        }
        if (curr > n) {
            return;
        }
        for (int dice = 1; dice <= 3; dice++) {
            printpath(n, curr + dice, ans + dice);
        }

    }

}
```

dice=1 2
PC3,3,21)
PC3,4,22)
PC3,5,23)
n=2
c=2
ans2
dice=1 2 3 4
PC3,1,1)
PC3,2,2)
PC3,3,3)
N=3
curr=0
ans=-
PPC3,0,-)



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        String ques = "abc";
        System.out.println("\n" + Print(ques, ""));
        // System.out.println("\n"+val);

    }

    public static int Print(String ques, String ans) {
        if (ques.length() == 0) {
            System.out.print(ans + " ");
            return 1;
        }
        char ch = ques.charAt(0);
        int x = Print(ques.substring(1), ans);// bc,""--> no
        int y = Print(ques.substring(1), ans + ch);// bc,"a"-->yes
        return x + y;

    }
```

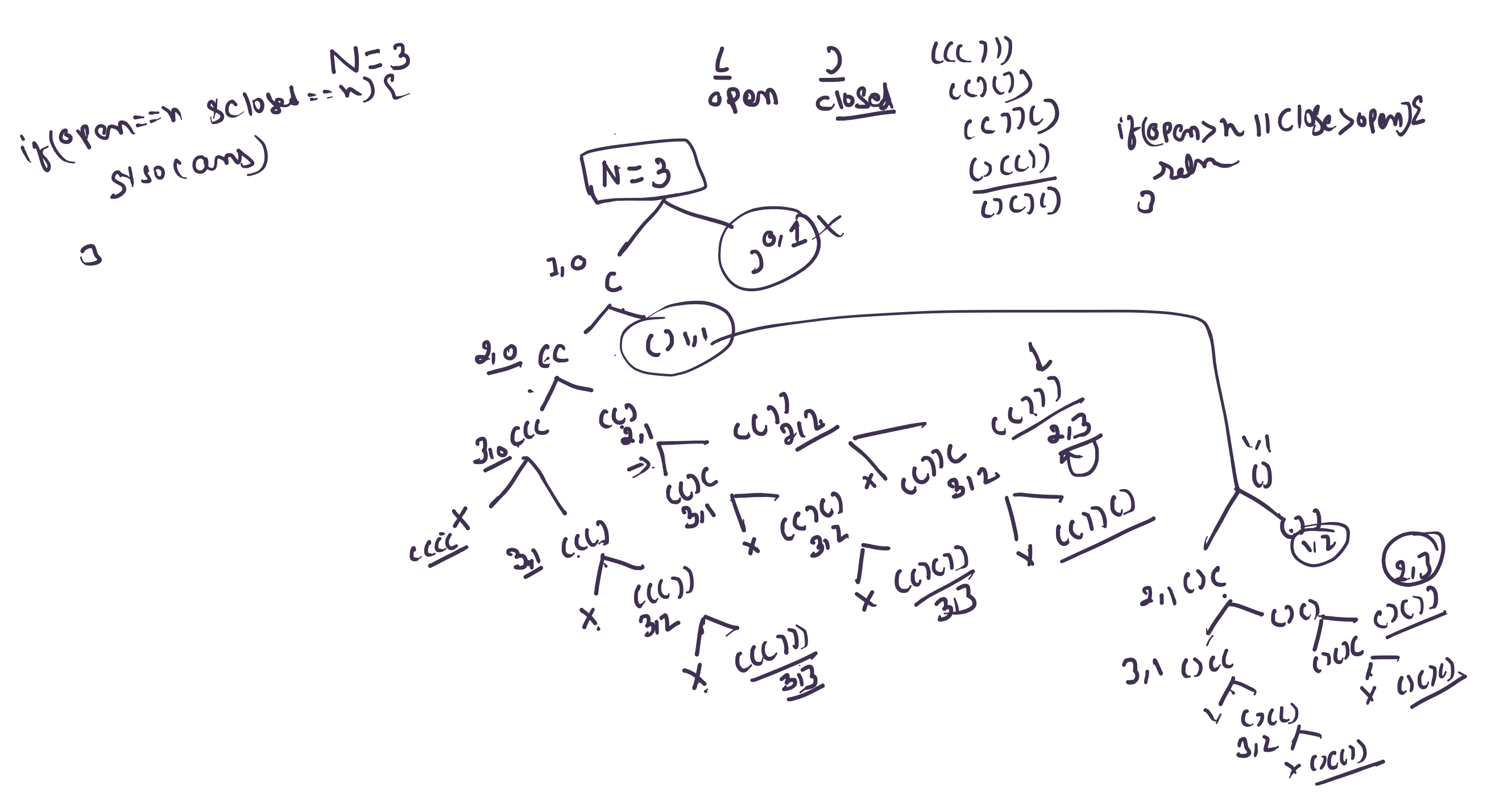
abc, -
bc, -
bc, a
c, -
c, b
ch='c'
x=P(-,a)
y=P(-,ac)
ch='b'
x=P(c,a)
y=P(c,ab)
ch=a
x=P(bc,-)
y=P(bc,a)
main
P(abc,-)
abc
bc
a
c

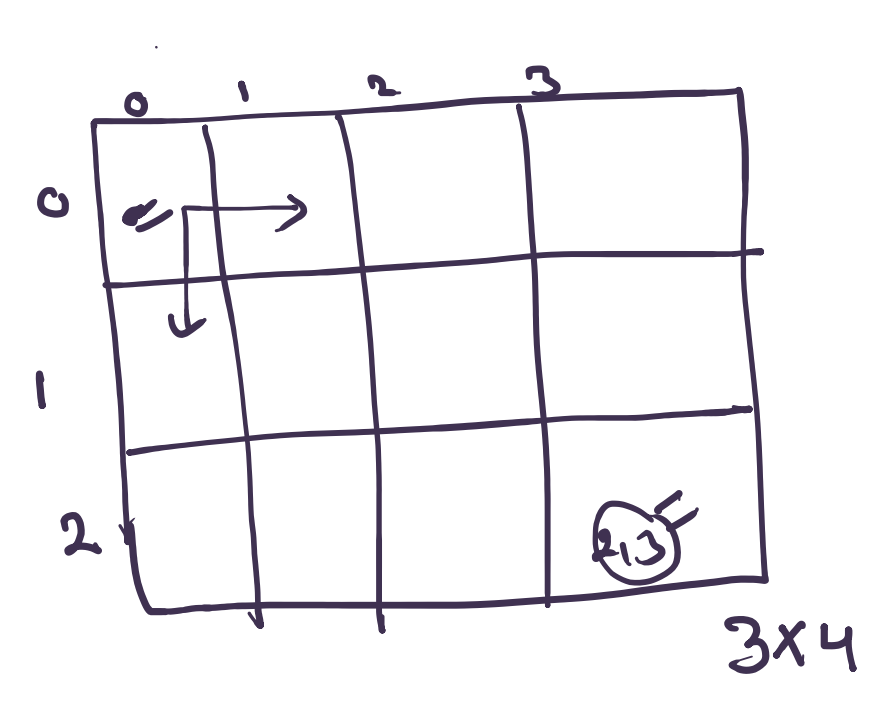


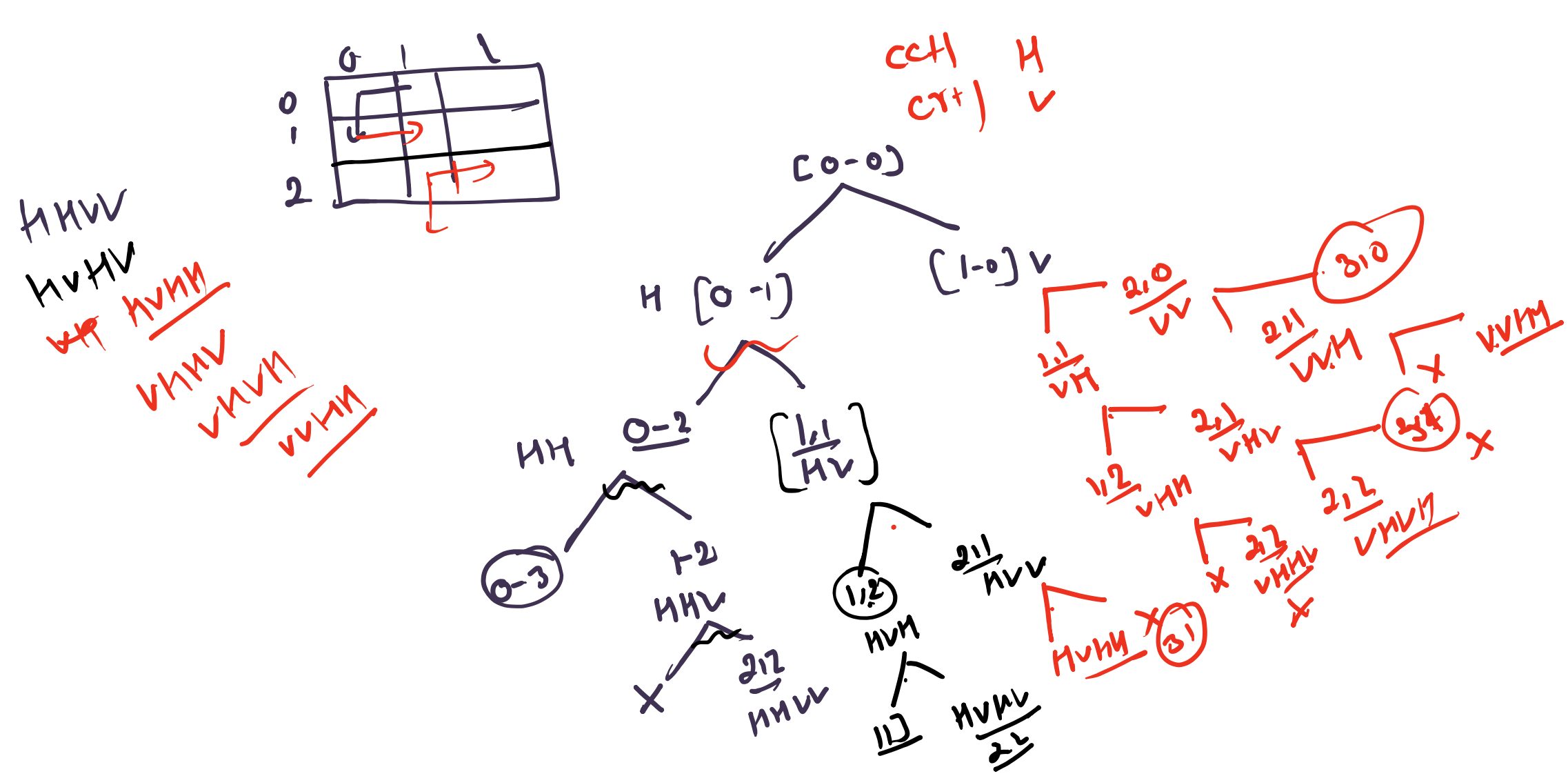
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        String ques = "abc";
        System.out.println("\n" + Print(ques, ""));
        // System.out.println("\n"+val);

    }

    public static int Print(String ques, String ans) {
        if (ques.length() == 0) {
            System.out.print(ans + " ");
            return 1;
        }
        char ch = ques.charAt(0);
        int x = Print(ques.substring(1), ans);// bc,""--> no
        int y = Print(ques.substring(1), ans + ch);// bc,"a"-->yes
        return x + y;

    }
```

8
abc, -
bc, -
bc, a
c, -
c, b
c, a
c, ab
4
2

---


N=3
if(open==n && closed==n){
Sysout(ans)
open
closed
((()))
(()())
(())()
()(())
()()()
if(open>n || close>open){
return
N=3


0
1
2
3
0
1
2
2,3
3x4


HHVV
HVHV
HVVH
VHHV
VHVH
VVHH
cc+1 H
cr+1 V
[0-0]
H [0-1]
[1-0] V
HH 0-2
1,1 HV
0-3
1,2
2,1
3,0